4. Differentiation		
4.1	Differentiation of e^{kx}, $\ln kx$, $\sin kx$, $\cos kx$, $\tan kx$ and their sums and differences.	
4.2	Differentiation using the product rule, the quotient rule and the chain rule.	Differentiation of $\operatorname{cosec} x$, $\cot x$ and $\sec x$ are required. Skill will be expected in the differentiation of functions generated from standard functions using products, quotients and composition, such as $2x^4 \sin x$, $\dfrac{e^{3x}}{x}$, $\cos x^2$ and $\tan^2 2x$.
4.3	The use of $\dfrac{dy}{dx} = \dfrac{1}{\left(\dfrac{dx}{dy}\right)}$	For example, finding $\dfrac{dy}{dx}$ for $x = \sin 3y$
4.4	Understand and use exponential growth and decay.	Students should be familiar with terms such as 'initial', 'meaning when' $t = 0$. Students may need to explore the behaviour for large values of t or to consider whether the range of values predicted is appropriate. Consideration of a second improved model may be required. Knowledge and use of the result $\dfrac{d}{dx}(a^x) = a^x \ln a$ is expected.